

Underwater object detection in images and videos

Jakub S. Zareba

219404347

Meng(Hons) Robotics Engineering with Year long work
placement

Meng Final Project Report

Academic Year 2025/26

Supervisor: Dr. Adrian Evans

Assessor: Dr. Dingguo Zhang

Number of Words in Abstract: 236

Number of Words in Main Body: 10,543

I have received ethical approval to pursue and complete this project.

1 Abstract

Underwater salient object detection presents unique challenges due to the severe and physically distinct image degradation inherent to underwater environments, including wavelength dependent light attenuation, contrast reduction and boundary blurring. Unlike terrestrial scenes, these degradations cannot be addressed through standard domain adaptation techniques, motivating investigation into whether image enhancement preprocessing can bridge the domain gap between terrestrial SOD models and underwater scenes without requiring full architectural redesign.

This dissertation evaluates five image enhancement methods spanning classical physics based approaches through to cutting edge deep learning, applied as preprocessing to a CNN based terrestrial SOD model retrained on the USOD10K benchmark dataset. Results are compared against TC-USOD, a specialised transformer based underwater SOD model, across 60 experimental conditions combining training time and inference time enhancement.

Key findings indicate that enhancement methods introducing strong distribution shifts improve SOD performance only when training and inference conditions are matched, whilst DehazeFormer, a transformer based terrestrial dehazing method, produced consistent improvements across conditions despite never being exposed to underwater imagery. Perceptual image quality metrics showed no reliable correlation with downstream SOD performance. CIRNet retrained on raw USOD10K data achieved performance within 1% of TC-USOD on F-MAX and S-MEASURE, suggesting the theoretical transformer advantage is largely unrealised at current underwater dataset scales.

The results conclude that architectural specialisation remains the primary performance determinant, but capable CNN based models with preprocessing represent a practical and competitive alternative for near term underwater SOD deployment [236 words (250 max)].

2 Table of Contents

Table of Contents

1 Abstract.....	2
2 Table of Contents.....	2
3 Table of Acronyms.....	4
4 Introduction & Context.....	5
4.1 SOD.....	5
4.2 Underwater Challenges.....	6
4.3 Domain Shift.....	7
4.4 SOD Architecture.....	8
4.5 Digital Image Enhancement.....	9
4.5.1 Classical Physics Based Methods.....	10
4.5.2 Deep Learning Based Methods.....	10
4.5.3 Enhancement as a Domain Adaptation Strategy.....	11
4.6 Research Questions.....	11
4.7 Evaluation Metrics.....	11
4.7.1 SOD Metrics.....	11

4.7.2 Enhancement Quality Metrics.....	12
4.7.3 The Metric Interaction.....	12
4.8 Report Structure.....	13
5 Methods and Results.....	13
5.1 Datasets.....	13
5.2 Experimental Setup.....	13
5.2.1 Model and Tool Modifications.....	14
5.2.2 Utility Scripts.....	14
5.2.3 Enhancement.....	14
5.2.4 SOD Model Training.....	15
5.2.5 Evaluation.....	15
5.3 Models.....	16
5.3.1 TC-USOD – Hybrid Transformer Baseline.....	16
5.3.2 CIRNet – CNN Based Model with Attention.....	16
5.3.3 Model Summary.....	17
5.4 Enhancement Methods.....	17
5.4.1 Dark Channel Prior – Discarded.....	17
5.4.2 DehazeFormer.....	18
5.4.3 FFA-Net.....	18
5.4.4 U-Shape Transformers.....	18
5.4.5 WWPF.....	19
5.5 Results.....	19
5.5.1 Enhancement Quality.....	19
5.5.2 Effect of Enhancement at Training.....	20
1 Enhancement at Training on Raw Images.....	20
2 Enhancement at Training with Matched Images.....	22
5.5.3 Effect of Enhancement at Inference.....	23
6 Discussion and Conclusions.....	24
6.1 Perceptual Quality Metrics and SOD Performance.....	24
6.2 Additive Nature of Domain Shifts.....	24
6.3 Effects of Architecture on Generalisation.....	25
6.4 Shortcomings of FFA-Net.....	25
6.5 Architecture Remains the Primary Performance Determinant.....	26
6.6 Practical and Deployment Considerations.....	27
6.7 Future Work.....	28
6.8 Conclusion.....	29
7 Project Review.....	30
7.1 What Went Well.....	30
7.2 Setbacks.....	30
7.3 Project Improvements.....	31
8 Acknowledgements.....	31
8.1 Use of Generative Artificial Intelligence.....	31
9 List of References.....	31
10 Appendices.....	34
10.1 Appendix A – Full Results Tables.....	34
10.2 Appendix B – DockerFiles.....	36
10.2.1 CIRNet.....	36
10.2.2 TC-USOD.....	37
10.2.3 DehazeFormer.....	37
10.2.4 U-Shape Transformer.....	37

3 Table of Acronyms

Acronym Definition

AUC	Area Under the ROC Curve
AP	Average Precision
CNN	Convolutional Neural Network
CMHA	Channel-wise Multi-Head Attention
CMSFFT	Channel-wise Multi-Scale Feature Fusion Transformer
CIRNet	Cross-modality Interaction and Refinement Network
DAM	Depth Auxiliary Module
DCP	Dark Channel Prior
DF	DehazeFormer
DL	Deep Learning
E-MAX	Maximum E-measure
E-MEAN	Mean E-measure
FFA	Feature Fusion Attention
F-MAX	Maximum F-measure
F-MEAN	Mean F-measure
GAN	Generative Adversarial Network
GT	Ground Truth
MAE	Mean Absolute Error
PSNR	Peak Signal to Noise Ratio
RGBD	Red Green Blue Depth
ROC	Receiver Operating Characteristic
SGFMT	Spatial-wise Global Feature Modelling Transformer

SOD	Salient Object Detection
SSIM	Structural Similarity Index Measure
TC-USOD	Transformer CNN Underwater Salient Object Detection
UCIQE	Underwater Colour Image Quality Evaluation
UIQM	Underwater Image Quality Measure
USOD	Underwater Salient Object Detection
ViT	Vision Transformer
WWPF	Weighted Wavelet Visual Perception Fusion

4 Introduction & Context

4.1 SOD

SOD is the task of identifying and isolating the most visually distinctive object in an image at pixel level precision [6]. As humans we do this naturally, differentiating backgrounds from points of interest without conscious effort. We can identify the dog in Figure 1 because it is visually distinct from its surroundings, not because it is a dog. SOD formalises this process computationally, producing binary masks that predict which pixels of an image represent the salient object. Interestingly Figure 1 also demonstrates the limitations of SOD, as the white part of the dog’s tail is not identified as part of the dog, but the right hand of the owner is.

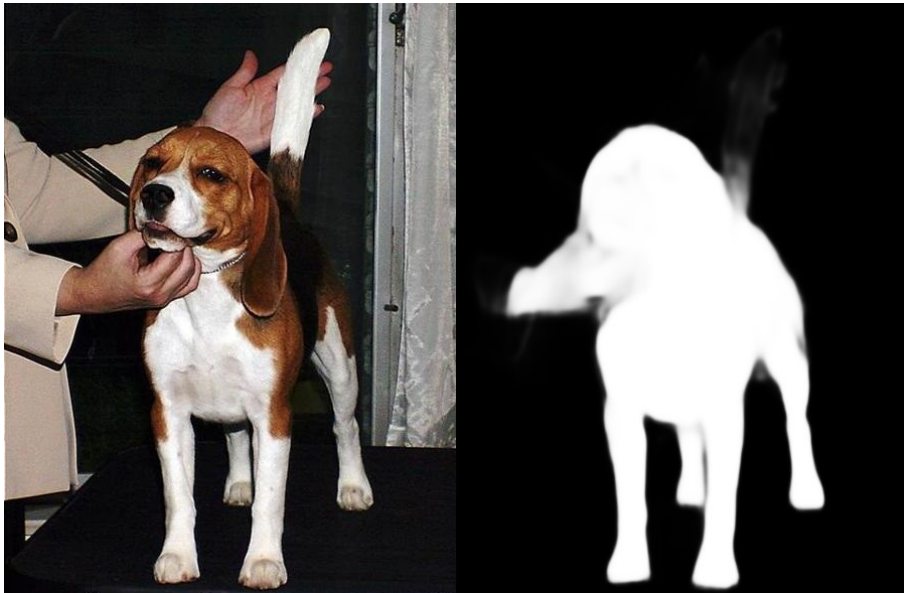


Figure 1: Example RGB image and corresponding predicted saliency map produced by CIRNet [1], from the NJU2K dataset [2].

SOD can then be used in tandem with object identification methods, such as bounding box detection [7], in order to reduce the computational overhead required when processing complex scenes. This can prove especially useful in context such as automatic image captioning [8], object tracking [9], visual question answering [10], person re-identification [11] and autonomous vehicles and robotics [12]. One of the first proposed SOD models was based on the center-surround mechanism, detecting spatial discontinuities across the colour, intensity and orientation channels [13]. This approach relied entirely on fixed computational operations with no learned parameters. The focus for many years after this was on improving a model’s ability to separate the foreground from the background using only the information available in the image, as explained by Borji [6]. Xu et al. [14] posed and successfully proved that semantic knowledge of an object has a significant impact on the performance of SOD. This can result in the same image producing two different prediction maps depending on the model’s training data, which would be directly impacted by the human annotator’s intent or preferences when deciding which object was salient [15].

Pure RGB based methods struggle however to differentiate salient objects in busy or low contrast scenes. The introduction of depth as an additional input channel marked a significant advancement for complex scene SOD, with RGBD based approaches fusing depth and RGB features to provide spatial context that purely RGB images cannot capture [16]. Figure 2 shows example images with their corresponding ground truth masks, depth maps and boundary maps, illustrating how depth complements colour information for saliency detection. Depth information is not always available at inference time however, motivating approaches that utilise depth during training without requiring it at deployment. Zhao et al. [12] demonstrated that depth used exclusively during training as a regularisation signal can produce models that surpass RGBD based methods at inference using only RGB input, suggesting depth provides structural priors that can be internalised without being present at inference time.

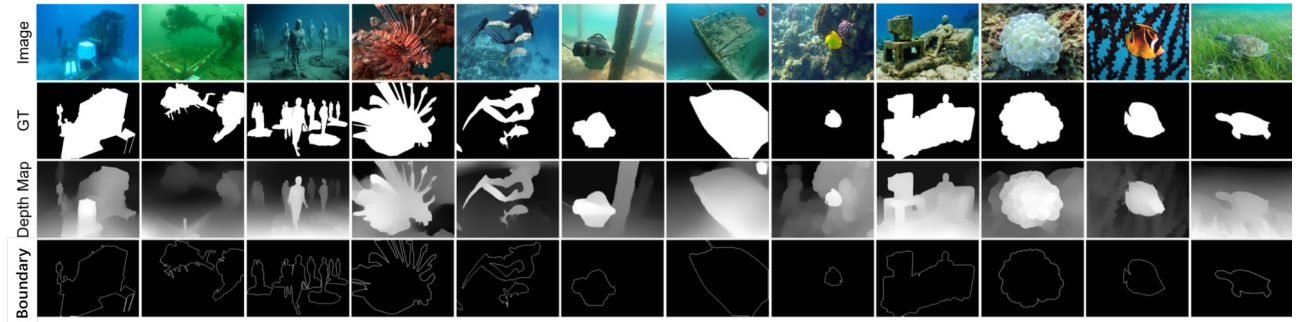


Figure 2: Multiple images with their corresponding GT, depth maps and boundary maps taken from USOD10K [3].

Initial SOD models based on purely computational methods began being replaced by DL approaches as early as 2015 [17], [18], [19], quickly demonstrating superior performance. These early models were subsequently expanded by integrating multi-scale feature aggregation, edge supervision, and attention mechanisms to improve performance on standard benchmarks [1], [20]. The introduction of vision transformers to SOD brought global self attention, capturing long range scene dependencies that convolutional architectures cannot model locally [21], and transformer based approaches now represent the state of the art for SOD across major benchmarks [16], [22].

4.2 Underwater Challenges

Underwater images suffer from multiple compounding degradations that terrestrial SOD models are not designed to handle. Red light attenuation is the most visually obvious. Water absorbs longer wavelengths of light faster than shorter ones [5], resulting in the red channel rapidly decreasing with depth whilst the blue and green channels persist. This is visible in Figure 3, where the heavy blue-green colour cast makes the fish and the background appear very close in colouration, but after enhancement with U-Shape Transformer the elements are more visually distinct.



Figure 3: Example image from the USOD10K dataset [3]. Left: raw underwater image. Right: enhanced output from U-Shape Transformer [4].

Light intensity drops rapidly with depth regardless of colour, as represented by Figure 4, reducing overall contrast in the scene. The edges of objects become blurred and poorly defined, which is a major issue for SOD models as boundary detection is central to accurate mask generation. The combination of these degradations removes many of the visual cues that SOD models rely on to generate accurate saliency masks.

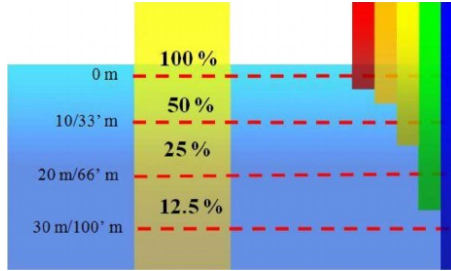


Figure 4: Representation of light intensity decrease relative to depth below the surface [5].

These challenges motivated a dedicated body of work in underwater SOD. Early approaches were limited by the scale of data available, with existing datasets totalling only a few thousand images [23], [24], [25], [26], making meaningful architectural comparisons and generalisations difficult to draw. Several approaches attempted to address the domain gap through architectural redesign, incorporating physics based priors and image enhancement methods directly into the network design [27], [28], [29], [30]. The release of USOD10K [3] marked a turning point for the field, providing the first dataset of a sufficient scale to make systematic comparison between approaches tractable. The USOD10K paper [3] also introduced TC-USOD, a specialised transformer based

USOD model developed and evaluated on the same dataset, providing both a large scale benchmark and a strong specialised baseline simultaneously.

4.3 Domain Shift

DL models learn to identify features that were present in their training data. When a model is deployed in an environment where the data has a different distribution, a different domain, performance degrades because the statistical patterns the model learned no longer match what it is being presented with [31]. This is domain shift and it represents one of the most significant practical limitations of DL based vision systems [32], [33], [34].

Domain shifts can arise from differences in imaging conditions, colour distribution, noise characteristics or scene content. In many cases the shift is relatively mild; a model trained on clear sky images applied during an overcast day, or a model trained on one camera sensor when used with another sensor. These shifts often don't change the distribution sufficiently to impact the effectiveness of DL computer vision, and as such can be addressed through data augmentation, fine tuning or domain generalisation techniques [35], [36]. These generalisation techniques function on the premise that the underlying data distributions are domain invariant and can be altered sufficiently to be approximated in the native domain, using functions such as Fast Fourier Transforms [35], [36].

The shift to underwater imagery is too significant due to the changes in environment affecting multiple channels in ways that are not replicated in terrestrial environments. The colour channel statistics are completely different due to the wavelength dependent attenuation being non-linear. The contrast and edge distributions are directly impacted by light absorption and scattering. The noise characteristics are the most similar to terrestrial, but the level of marine snow [37] is significantly heavier than most land environments.

Beyond the imaging conditions, the subject matters themselves are effectively alien to any terrestrial model. The shapes of the flora, fauna and landscape are not represented in terrestrial habitats. Underwater salient objects include organisms with radially symmetric or amorphous geometries (octopuses, jellyfish, coral formations) that bear no resemblance to the objects dominating terrestrial SOD datasets. Using a model that was trained on terrestrial data it is easy to see how there would be a major issue with such a model trying to interpret underwater images. FIGURE XXX shows how these challenges aggregate into significant failure of an SOD model.

Domain generalisation techniques alone are insufficient to bridge the terrestrial to underwater gap. The shift is not a matter of distribution drift that can be approximated through augmentation or fine tuning. It requires complete retraining on underwater data to account for the geometric and structural differences in underwater subjects. Even then, retraining alone cannot overcome the fundamental differences in how light behaves underwater, which systematically alters the colour, contrast and edge characteristics that SOD models rely on. Rather than adapting the model to the target domain, this dissertation investigates an alternative strategy that is often used in tandem with architectural changes; transforming the target domain images toward the source domain through image enhancement preprocessing, reducing the domain gap at the input level rather than the feature level.

4.4 SOD Architecture

Following the success of AlexNet [38], CNNs dominated computer vision for nearly a decade. They learn hierarchical features through local convolutional operations that encode spatial locality and translation invariance. These built in assumptions, inductive biases, allow CNNs to learn effectively from relatively small datasets because the architecture already contains strong priors about local feature relationships.

The introduction of vision transformers [21] challenged this dominance by demonstrating that self attention applied to image patches could exceed CNN performance on classification tasks at scale. Rather than processing local neighbourhoods through fixed kernel operations, transformers model relationships across all image patches simultaneously, capturing global context that CNNs can only approximate through additional DL layers or explicit multi-scale aggregation. For SOD especially this global context is valuable, identifying which object in a scene is salient requires understanding the entire scene, not just the local features.

This transformer advantage comes with a critical trade-off; transformers do not have the inductive biases of CNNs, meaning that all spatial relationships must be learnt from training data. This drastically increases their demand for large and varied datasets. In the original ViT paper it is explicitly noted that when trained on modest datasets, CNNs continue to outperform ViTs [21]. Performance on vision tasks has been shown to scale logarithmically with training data volume [20], and this relationship is particularly evident in transformer based approaches. The lack of inductive biases requires a much larger training volume to learn the same structural features as CNNs.

The availability of large scale SOD datasets renders this dependency largely irrelevant. Datasets of hundreds of thousands of images provide sufficient volume that ViTs can converge effectively and achieve their theoretical performance. USOD faces a major limitation in this regard. USOD10K at 10,255 images is the largest USOD dataset in existence but is modest when compared to its terrestrial counterparts, approximately 33 times smaller than COCO [39] and 185 times smaller than Open Images [40]. These are the datasets that transformers are typically trained on.

This constraint has a direct empirical consequence that is investigated in this dissertation. At the data scales realistic for underwater SOD, the theoretical transformer advantage may be largely unrealised; CNNs have not been pushed to their performance ceiling at this scale, and their stronger inductive biases may partially compensate for the data deficit. This hypothesis is tested in section 5.

CNNs also retain practical advantages that benchmark comparisons often omit. Inference speed, memory efficiency and deployability on limited hardware remain significantly better for CNN based approaches. For real world underwater deployment on autonomous underwater vehicles or remotely operated vehicles these constraints are critical and directly impact design choices. A transformer SOD model that achieves marginally higher results on benchmarks but cannot run in real time is less desirable when hardware limitations are a major concern. This becomes more apparent as the resolution of an image increases, with CNNs exhibiting $O(n)$ time where transformers exhibit $O(n^2)$ time complexity [41] with the best case scenario dropping to $O(n \log n)$ in recent developments [42].

The question this poses for USOD is therefore not which architecture performs better in ideal conditions, but which is most viable, practical and improvable given the data and hardware constraints that exist within this domain. This is the context that inspired the question of this dissertation, whether the underwater domain is developed enough to support a shift to transformer based architecture.

4.5 Digital Image Enhancement

Image enhancement covers a broad range of methods designed to improve the perceived quality of degraded images by restoring lost information, correcting colour distortion and improving contrast and edge definition. For underwater imagery specifically, enhancement addresses the physical degradations discussed in section 4.2, red channel attenuation, contrast reduction and the loss of edge definition, all of which impact SOD performance. It is important to note that the enhancement is judged on improving the image quality for human sight, not necessarily for preprocessing for computer vision.

4.5.1 Classical Physics Based Methods

Simpler enhancement approaches that are often attempted initially are based on the physical model of light propagation. The atmospheric scattering model describes how light interacts with suspended particles between a scene and a camera, which was exploited by Dark Channel Prior [43]. DCP was developed by observing that in most non-haze patches at least one colour channel has very low intensity. This prior enabled transmission map estimation and haze-free image recovery without any learned parameters, establishing a baseline that is frequently used as a comparison in following enhancement works.

Physics based methods have clear practical advantages: they require no training data, are computationally lightweight and are immediately deployable without any dataset collection or model training. They are however severely limited due to the fixed assumptions about their physical environment that are not universal.

DCP’s assumptions about atmospheric scattering require that degradation be relatively uniform and depth dependent in a predictable way. Underwater environments violate these assumptions. The degradation is wavelength dependent in a non-linear way, varies with water turbidity and salinity, all of which differ significantly based on geography. A physics based model optimised for one body of water will produce meaningfully different results in any other body of water where the same conditions don’t persist. This geographic variance means that physics based enhancement cannot generalise across the diverse conditions present in real world data.

More sophisticated physics based methods such as WWPF [44] address some of these limitations by incorporating wavelet decomposition and attenuation map guided colour correction, adopting priors more representative of underwater light propagation models. However even this method relies on assumptions about colour channel relationships and contrast distribution that may not hold consistently across diverse underwater environments.

4.5.2 Deep Learning Based Methods

The limitations of fixed physical priors motivated the development of deep learning based enhancement methods that learn from diverse data rather than encoded assumptions. These methods generalise better across environments because they adapt to the patterns present in training data rather than assuming universal physical behaviour, making them more suitable for the variance of conditions encountered in real underwater datasets.

Supervised deep learning methods do share a fundamental limitation for underwater applications; they require paired degraded and clean training images. For atmospheric haze this is achievable through synthetic haze generation on clean images. In underwater environments no haze-free ground truth exists. The physical processes that cause underwater degradation cannot be isolated or manipulated in any way that could generate clean reference images. When considering the domain shift constraints discussed in 4.3, using terrestrial images that were synthetically degraded in a manner reminiscent of underwater environments is not a viable solution. This constraint limits the development and training of deep learning models, as LSUI [4] is considered a large underwater paired image dataset, even though it contains approximately 4000 images, a scale modest by most supervised learning standards.

4.5.3 Enhancement as a Domain Adaptation Strategy

Despite the extensive body of work in underwater image enhancement and the parallel work on USOD, the specific contributions of enhancement to bridging the domain shift from terrestrial to underwater has not been systematically evaluated. Enhancement methods have been incorporated as components within jointly trained end to end systems [27], but never isolated as an independent variable. In terrestrial computer vision, de-hazing methods have been shown to have limited or negative effects on high level vision tasks including classification and detection, despite demonstrably improving image quality [45]. This counter-intuitive finding motivates the question of whether the same effect holds for SOD in underwater scenes, where the degradation actively suppresses the edge and contrast information that SOD models rely on, rather than simply adding noise on top of otherwise clean features. The enhancement can also be applied at either training or inference or both. There is also the question of whether any terrestrial models can be used in underwater environment or whether architectural changes are the only solution. Whether terrestrial enhancement models transfer sufficiently to underwater preprocessing without domain specific retraining, and whether the type of enhancement method matters as much as its presence, remain open questions this dissertation addresses directly.

4.6 Research Questions

Substantial bodies of work exist independently in underwater image enhancement and USOD, covering approaches ranging from classical physics based enhancement methods, deep learning enhancement methods, CNN based SOD models and ViT based SOD models. The interaction of these approaches however has not been explored for the purpose of domain adaptation or performance improvement. This leaves open the question of whether modular decoupled enhancement alone can bridge the domain gap between terrestrial and underwater SOD, without joint training or architectural redesign. This dissertation aims to determine whether image

enhancement can close this gap, and whether the performance of CNN based SOD models can be made competitive with ViT based approaches under realistic underwater data constraints.

4.7 Evaluation Metrics

4.7.1 SOD Metrics

SOD model performance is evaluated using a standard suite of metrics that measure the accuracy of predicted saliency masks against ground truth annotations. These metrics are well established in the SOD literature and allow direct comparison against published results.

MAE measures the average pixel level difference between the predicted saliency map and the ground truth mask, providing an overall measure of prediction accuracy. Lower MAE indicates better performance.

F-measure is the harmonic mean of precision and recall, measuring how well the predicted mask captures the salient region without including background pixels. Both maximum F-measure, computed at the optimal threshold, and mean F-measure, averaged across all thresholds, are reported to provide a complete picture of detection performance across operating conditions.

E-measure captures both pixel level accuracy and image level semantic alignment between predicted and ground truth masks, providing a more holistic assessment than pixel level metrics alone. Maximum and mean E-measure are reported.

S-measure evaluates the structural similarity between predicted and ground truth saliency maps, measuring both object level and region level structural fidelity. It is particularly informative in complex scenes where structural accuracy matters more than pixel accuracy.

AP summarises performance across the full precision recall curve, and AUC measures the model's ability to rank salient pixels above background pixels across all possible thresholds. Together these provide a complete characterisation of model performance independent of any single threshold choice.

4.7.2 Enhancement Quality Metrics

Evaluating enhancement method output quality presents a significant challenge in the underwater domain. Standard reference based metrics such as PSNR and SSIM require a clean ground truth image to compare against. In underwater environments no haze free ground truth exists; the physical degradation cannot be reversed to produce a reference, and paired datasets such as LSUI [4] contain only approximately 4000 images, insufficient to represent the full diversity of underwater conditions. Reference based metrics are therefore not viable for evaluating enhancements on USOD10K.

This is significant in itself. The same absence of paired data that constrains supervised enhancement model training also prevents extensive quantitative evaluation of enhancement output quality using standard metrics. This results in no-reference evaluation metrics being more desirable and applicable in the underwater domain.

UCIQE [46] is a no-reference metric specifically designed for underwater colour images, measuring a linear combination of chroma, saturation and contrast in the CIELab colour space. It was developed through subjective testing to correlate with human perception of underwater image quality, capturing the non-uniform colour cast, blurring and low contrast characteristics of underwater imagery. Unlike atmospheric image quality metrics it accounts for the specific degradation profile of underwater scenes.

UIQM [47] is a complementary no-reference underwater metric combining measures of colour, sharpness and contrast, providing a broader assessment of perceptual image quality than UCIQE.

Both metrics are used to evaluate the enhancement methods' outputs independently of SOD performance, allowing direct comparison between the perceptual quality improvements and the downstream effects on SOD metrics. A key question this dissertation addresses is whether these two measures are correlated; whether enhancement methods that score highest on UCIQE and UIQM also produce the best SOD performance, or whether perceptual quality and detection utility are decoupled.

4.7.3 The Metric Interaction

It is worth noting explicitly that the metrics used to evaluate enhancement methods and those used to evaluate SOD models are entirely disjoint. Enhancement literature evaluates on PSNR, SSIM, UCIQE and UIQM. SOD literature evaluates on MAE, F-measure, S-measure, AP and AUC. No shared metric exists that directly measures whether enhancement improves detection. This disconnect is precisely why the question this dissertation investigates has not been systematically addressed. The two fields have been optimising for different objectives with no common ground truth to connect them.

4.8 Report Structure

This report is structured as follows. Section 4 introduces the context and background of the work, reviewing relevant literature across underwater salient object detection, image enhancement, domain shift, and CNN versus transformer architectures, culminating in the research questions this dissertation addresses. Section 5 presents the methodology and results, describing the experimental design, models, enhancement methods, and evaluation metrics before presenting results organised around the stated research questions. Section 6 discusses the findings and draws conclusions regarding the viability of image enhancement as a domain adaptation strategy for underwater SOD. Section 7 reviews the project process. Section 8 contains acknowledgements including a statement on the use of generative AI. Full references and appendices follow.

5 Methods and Results

5.1 Datasets

All training and evaluation in this dissertation uses USOD10K [3], the first large scale USOD dataset, introduced alongside TC-USOD in 2025. USOD10K provides three predefined splits:

- TR – 7,178 images used for all model training

- VAL – 2,051 images used as the secondary evaluation set
- TE – 1,026 images used as the primary evaluation set

The TR and VAL splits provide RGB images, ground truth saliency masks, edge maps and depth maps. The TE split provides RGB images, ground truth masks and depth maps. All models are evaluated on both TE and VAL to provide a robust performance picture across different image distributions within the dataset.

The predefined splits are used without modification to ensure comparability with results reported in the original USOD10K paper and subsequent benchmark work.

5.2 Experimental Setup

All models were containerised using Docker to ensure reproducible environments and manage dependency conflicts between models. TC-USOD and CIRNet share compatible dependencies, however separate containers were used for clarity. DehazeFormer and U-shape transformer required separate docker environments due to minor library incompatibilities. Where no specific library version was required the most recent stable versions of the packages were used.

5.2.1 Model and Tool Modifications

All of the models required minor modifications to run in the project environment. Many of these were due to the paths and file names being hard-coded. To ensure reproducibility the changes are listed here.

TC-USOD: Changed test save path ending to `‘/USOD10K/’`. Changed test path to `‘/workspace/data/USOD10k/TE’`.

UC-Net: UC-Net was initially considered as an additional CNN baseline. Path variables were updated to absolute paths and deprecated `misc.imsave` calls were replaced with `PIL.Image.save`. Despite these fixes UC-Net failed during training on three separate attempts, terminating at 8, 15 and 23 epochs respectively with NaN tensor outputs. The root cause was not resolved within the available time and UC-Net was discarded from the experiment. This is discussed further in section 7.

CIRNet: State dict loading had the `‘module.’` prefixes removed for compatibility with single GPU operation. `‘torch.load’` was updated to include `‘weights_only=False’` to allow using the pretrained model. `‘gpu_id’` default was changed from `‘6,7’` to `‘0’` to use a single GPU. Test dataset paths were updated as necessary to use the correct dataset variants.

FFA-Net: `‘torch.load’` updated to include `‘weights_only=False’`.

U-Shape Transformer: Input path updated in `‘test.ipynb’`.

WWPF: Input and output directories updated in `‘WWPE.m’`. File extensions changed from `‘.jpg’` to `‘.png’` for consistency with USOD10K. Output file name suffix removed to match expected naming convention.

5.2.2 Utility Scripts

One utility script was written from scratch to resize images from U-Shape Transformer, ‘resize.py’

5.2.3 Enhancement

The GitHub repositories for each enhancement method were cloned and run using their default parameters and pretrained weights. MVG-FD [48] was considered but excluded as no pretrained weights were available, falling outside the scope of this dissertation which evaluates existing pretrained enhancement methods rather than training new ones.

For each enhancement method, the RGB images from the TR, VAL and TE splits of USOD10K were parsed through the respective model using default settings. All other modalities (ground truth masks, depth maps and edge maps) were left entirely unchanged. The enhanced RGB outputs replaced the original RGB folder for each dataset variant, producing four preprocessed versions of USOD10K alongside the original:

- USOD10K_Base - original unmodified RGB images
- USOD10K_De hazeFormer – RGB processed by De hazeFormer
- USOD10K_FFA_Net – RGB processed by FFA-Net
- USOD10K_U_Shape – RGB processed by U-Shape Transformer
- USOD10K_WWPF – RGB processed by WWPF

For De hazeFormer, FFA-Net and U-Shape Transformer, which have evaluation scripts built directly into their test pipelines, the input RGB images were also used as ground truth references. Since USOD10K does not provide haze-free ground truth images and reference based metrics such as PSNR and SSIM were not used in this dissertation, this had no impact on the validity of the outputs. Images were processed sequentially with input and output directories updated manually between runs rather than via command line arguments.

Each preprocessed dataset was subsequently evaluated using UCIQE and UIQM to confirm meaningful output changes. During this evaluation DCP was found to have completely failed; processed images contained large solid blocks of colour rendering them unusable, consistent with the known incompatibility between atmospheric scattering priors and underwater imagery discussed in section 4.5.1 DCP was therefore discarded entirely.

5.2.4 SOD Model Training

TC-USOD was cloned from its official GitHub repository and retrained on USOD10K TR following repository instructions, with the sole modification of setting epochs to 100. TC-USOD was trained only on the base unmodified USOD10K TR images.

CIRNet was cloned from its official GitHub repository. To avoid output naming conflicts between training runs, CIRNet was copied into five separate directories, one per dataset variant. CIRNet was retrained on the following datasets:

- USOD10K_Base

- USOD10K_De hazeFormer
- USOD10K_FFA_Net
- USOD10K_U_Shape
- USOD10K_WWPF

The pretrained terrestrial CIRNet weights were kept in the original directory as a sixth model for domain shift baseline evaluation.

All models were trained for 100 epochs using default hyperparameters from their respective codebases, with the sole exception of epoch count which was standardised across all models to ensure fair comparison. Training was performed on an AMD Ryzen 5 9600X CPU with an RX 9060 XT 16GB GPU and 32GB DDR5 6000MHz RAM. The AMD ROCm backend was used to allow GPU acceleration. TC-USOD training completed in approximately 22.6 hours. Each CIRNet training run took approximately 39.2 hours. With 5 CIRNet training runs this totalled approximately 219 hours across all six models.

5.2.5 Evaluation

The evaluation script from TC-USOD was adapted for use with CIRNet outputs. Components specific to TC-USOD's output format were removed to allow it to run on CIRNet generated masks while maintaining consistent metric computation across all models. This ensured all SOD metrics were computed using an identical evaluation pipeline regardless of which model generated the predictions.

All trained models were evaluated on both the TE and VAL splits of all five USOD10K variants at inference time, in addition to the training time preprocessing already incorporated into the retrained CIRNet models. This produces a total of 60 unique model and image condition combinations across the full experiment.

5.3 Models

5.3.1 TC-USOD – Hybrid Transformer Baseline

TC-USOD [3] was introduced alongside USOD10K in the same paper, developed and evaluated by the same research group on the same dataset, making it a natural specialised baseline for this experiment.

TC-USOD adopts a hybrid architecture consisting of a transformer encoder and a convolutional decoder. The encoder uses T2T-ViT as its backbone, applying token-to-token transformations to progressively tokenise the input and model local structure information through restructurisation and soft split operations, capturing global scene context through self-attention. The decoder is convolutional, using multi-level feature fusion and multi-task learning to generate full resolution saliency maps and object boundaries simultaneously. A depth auxiliary module with cross-modality fusion is used to incorporate depth map information, purifying depth features through channel and spatial attention before fusing them with RGB features. Training uses a hybrid loss combining BCE,

IoU, Dice and SSIM losses. TC-USOD is therefore primarily transformer based in its feature extraction but relies on convolutional operations for decoding and prediction.

TC-USOD is trained once on base USOD10K TR images with no preprocessing applied, using the default hyperparameters as defined in the argparser. At inference it is evaluated on all five variants of USOD10K TE and VAL. This tests what effects enhancement at inference time, on a model that has never seen enhanced images, produces.

5.3.2 CIRNet – CNN Based Model with Attention

CIRNet [1] was chosen as the CNN based comparison model. It is an end-to-end three stream encoder-decoder network for RGBD SOD, processing RGB, depth and RGBD modalities through separate but interacting streams. CIRNet incorporates cross-modality interaction through a progressive attention guided integration unit in the encoder stage and an importance gated fusion unit in the decoder stage, as well as a refinement middleware containing self-modality attention refinement and cross-modality weighting refinement units. These attention mechanisms capture long range dependencies between modalities in a manner that approximates the transformer global self attention behaviour, whilst remaining fundamentally convolutional.

This architectural characteristic was a deliberate selection criterion. TC-USOD incorporates a convolutional decoder alongside its transformer encoder, suggesting that the inductive biases of convolution, specifically spatial locality and translation invariance, remain valuable for precise boundary delineation even when global context is captured by the encoder. CIRNet's attention mechanisms operate over smaller regions than TC-USOD's global self attention, but if performance remains comparable this would suggest that global context is less critical than the ability to detect features consistently regardless of their spatial position. Crucially, both architectures combine attention and convolutional components, differing primarily in the scope of their attention and the primacy of each mechanism rather than in fundamental architectural philosophy. This makes the comparison more controlled and informative than a simple CNN versus transformer binary, isolating the contribution of attention scope and architectural orientation rather than confounding the comparison with entirely different design paradigms.

CIRNet was originally trained on terrestrial RGBD SOD datasets. The pretrained terrestrial weights are retained as a domain shift baseline, evaluated directly on USOD10K without retraining to quantify the raw performance penalty of applying a terrestrial model to underwater images without adaptation. Five additional CIRNet models are retrained on each preprocessed USOD10K variant, allowing direct comparison of enhancement method effects on training data distribution.

At inference all 6 CIRNet models are evaluated on all 5 USOD10K variants on both TE and VAL.

5.3.3 Model Summary

Model Name	Base Model	Retrained	Training Domain	Training Enhancement	Epochs	Hyperparameters
TC-USOD	TC-USOD	Yes	Underwater	None	100	Default
CIRNet	CIRNet	No	Terrestrial	N/A	N/A	Default
CIRNet Base	CIRNet	Yes	Underwater	None	100	Default
CIRNet DehazeFormer	CIRNet	Yes	Underwater	DehazeFormer	100	Default
CIRNet FFA-Net	CIRNet	Yes	Underwater	FFA-Net	100	Default
CIRNet U-Shape	CIRNet	Yes	Underwater	U-Shape Transformer	100	Default
CIRNet WWPF	CIRNet	Yes	Underwater	WWPF	100	Default

Table 1: Summary of choices made during model training

5.4 Enhancement Methods

5.4.1 Dark Channel Prior – Discarded

DCP was the first enhancement method evaluated. It estimates a transmission map using the observation that in most non-haze patches at least one colour channel has very low intensity. This prior was developed for atmospheric scattering conditions and assumes a relatively uniform depth dependent degradation model.

When USOD10K was parsed through DCP, the outputs were completely unusable. Rather than producing degraded or low quality enhancement, DCP destroyed image content entirely. Large portions of images were replaced with solid blocks of colour. This indicates that the underwater colour channel statistics are so far removed from DCP’s core assumptions that transmission map estimation fails completely rather than producing inaccurate results. DCP was therefore discarded from the experiment and no SOD models were trained on DCP processed images.

5.4.2 DehazeFormer

DehazeFormer [49] is a transformer based single image dehazing method built on a modified Swin Transformer architecture. It applies windowed self attention with parallel convolution to capture both local and global context in transmission map estimation, incorporating several modifications to standard transformer design, including a custom normalisation layer and activation function, specifically to improve performance on low level vision tasks such as dehazing. It was trained on the RESIDE dataset and achieves state of the art performance on standard atmospheric haze benchmarks.

DehazeFormer is included to test whether transformer based preprocessing transfers any architectural advantage to a CNN based detection model. Its windowed attention mechanism also

provides an interesting middle ground between CIRNet's local attention and TC-USOD's global attention, adding a further point of comparison on the attention scope spectrum.

5.4.3 FFA-Net

FFA-Net [50] is a supervised dehazing method that uses feature fusion attention to perform end-to-end haze removal. It is one of the most widely cited dehazing methods in the literature and is commonly used as a benchmark for atmospheric haze removal. It was trained on the RESIDE dataset and has never been exposed to underwater imagery.

FFA-Net is included as a well validated community baseline, representing the performance of a strong terrestrial supervised dehazing model applied directly to underwater preprocessing without any domain specific adaptation. This tests whether the abundance of terrestrial haze training data can be leveraged to improve underwater SOD without underwater specific retraining.

5.4.4 U-Shape Transformers

U-Shape Transformer [4] is an underwater specific image enhancement method that explicitly models the inconsistent attenuation characteristics of underwater images across both colour channels and spatial regions. It was introduced alongside the LSUI dataset [4], containing 4,279 paired underwater images across diverse underwater scenes, water types and lighting conditions.

The architecture uses two purpose built modules: a Spatial-wise Global Feature Modelling Transformer (SGFMT) to model global spatial features and reinforce attention on severely attenuated regions, and a Channel-wise Multi-Scale Feature Fusion Transformer (CMSFFT) to perform channel-wise attention across multi-scale features, directing the network's attention toward the more severely attenuated colour channels. A multi-colour space loss function combining RGB, LAB and LCH colour spaces is used during training to improve colour restoration accuracy. Together these components directly address the red channel attenuation and spatially non-uniform degradation characteristic of underwater imagery discussed in section 4.2.

U-Shape Transformer represents the domain specific deep learning baseline in this experiment, a purpose built underwater enhancement method trained on underwater paired data. Its inclusion allows direct comparison between terrestrial enhancement methods applied to underwater images and a method specifically designed and optimised for the underwater degradation model.

5.4.5 WWPF

WWPF [44] is a physics informed underwater image enhancement method that operates without deep learning training, combining three sequential processing stages to address the specific degradation characteristics of underwater imagery.

The first stage applies attenuation map guided colour correction, compensating colour channels relative to a luminance reference channel to address the non-uniform colour cast caused by wavelength dependent light attenuation. The second stage applies both global and local contrast enhancement. Global contrast is optimised using maximum information entropy to determine the best dynamic range threshold, while local contrast is enhanced using an integral image approach to efficiently compute local block mean values and adaptively amplify high frequency components.

The third stage applies weighted wavelet visual perception fusion, using first order wavelet decomposition to separate global and local contrast enhanced images into approximate low frequency and vertical, horizontal and diagonal high frequency components across a four layer pyramid, then reconstructing a final enhanced image using gradient based weighting factors that emphasise the most detail rich frequency components.

Unlike the deep learning methods WWPF requires no training data and no pretrained weights, making it immediately applicable to any underwater dataset without domain specific adaptation. It also demonstrates generalisation to other low visibility conditions including fog and low light without parameter adjustment, suggesting its physics informed design captures general visibility degradation properties rather than underwater specific patterns alone.

WWPF is included alongside U-Shape Transformer to compare physics informed and learned approaches to underwater specific enhancement, testing whether the additional complexity of a deep learning model produces meaningful downstream SOD improvements over a well designed parameter free alternative.

5.5 Results

5.5.1 Enhancement Quality

Prior to evaluating downstream SOD performance, the output quality of each enhancement method was assessed using UCIQE and UIQM on all three USOD10K splits. These no-reference metrics provide an independent measure of perceptual image quality improvement, allowing subsequent analysis to determine whether methods that produce higher quality enhanced images also produce better SOD performance.

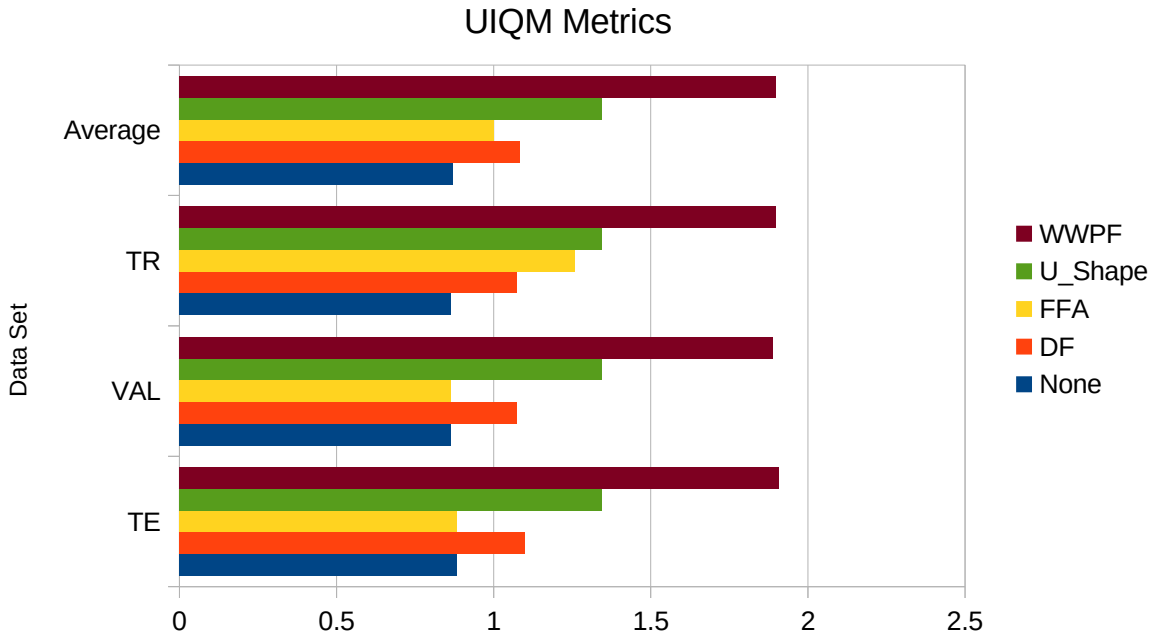


Figure 5: UIQM results of all of the datasets used.

WWPF achieved the highest score on both UCIQE and UIQM across all three splits, with average UCIQE of 0.5965 and UIQM of 1.8982. U-Shape transformer performed second on both metrics, with UCIQE of 0.5583 and UIQM 1.3453. DehazeFormer achieved a UCIQE of 0.5432 and UIQM of 1.0834. FFA-Net performed the worst of the enhancement methods with UCIQE of 0.5379 and UIQM of 1.0016, representing minimal and moderate improvement over the unenhanced baseline of 0.5332 and 0.8705 respectively.

5.5.2 Effect of Enhancement at Training

1 *Enhancement at Training on Raw Images*

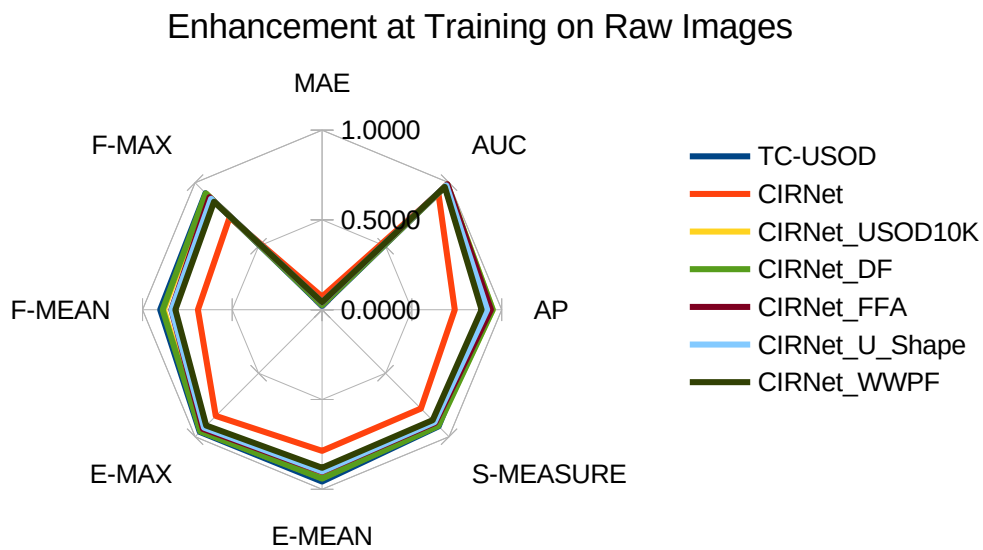


Figure 7: Radar graph of re-trained models on raw USOD10K images.

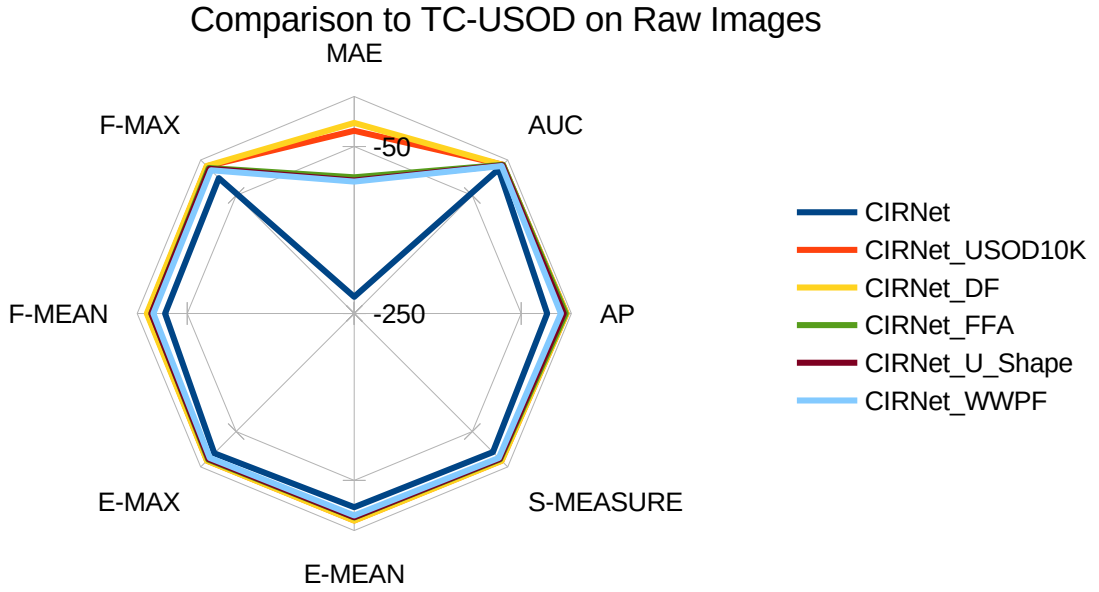


Figure 8: Radar graph showing performance relative to TC-USOD.

Figure 7 and Figure 8 present the performance of all retrained CIRNet models evaluated on unenhanced USOD10K images, averaged across both the TE and VAL splits, with TC-USOD as the reference baseline.

CIRNet retrained on raw USOD10K images showed substantial improvement over the terrestrial pretrained weights across all metrics, confirming the expected domain shift penalty of applying a terrestrial model directly to underwater images without adaptation. CIRNet_DF achieved the closest performance to TC-USOD of all retrained variants, with MAE of 0.0267 compared to TC-USOD's 0.0219, and F-MAX of 0.9154 compared to 0.9177, a difference of less than 0.3%.

CIRNet_USOD10K performed comparably, with MAE of 0.0287 and F-MAX of 0.9123.

CIRNet_FFA, CIRNet_U_Shape and CIRNet_WWPF all performed below CIRNet_USOD10K on raw images, with FFA producing MAE of 0.0410 and F-MAX of 0.8858, suggesting that training on FFA-Net enhanced images actively degraded performance on unenhanced images relative to training on raw data alone. AUC and AP showed high scores with minimal variation across all models and are discussed further in section 6.

2 Enhancement at Training with Matched Images

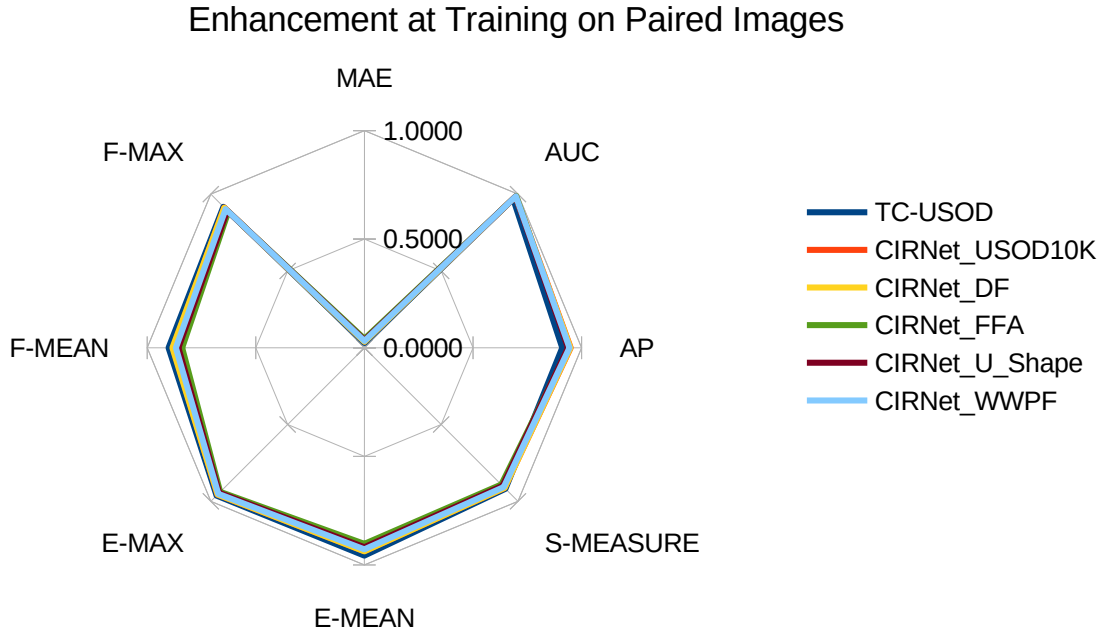


Figure 9: Radar graph presenting a comparison of all retrained models in their ideal conditions.

Figure 9 present all retrained CIRNet models evaluated on their matched enhanced image sets alongside TC-USOD as the reference baseline.

CIRNet_WWPF and CIRNet_U_Shape showed the largest improvements under matched conditions relative to their performance on raw images, with CIRNet_WWPF MAE improving from 0.0421 to 0.0312 and F-MAX improving from 0.8499 to 0.9028, and CIRNet_U_Shape MAE improving from 0.0418 to 0.0346 and F-MAX from 0.8741 to 0.8925. CIRNet_DF showed minimal change between raw and matched conditions, suggesting DehazeFormer enhanced images are sufficiently similar in distribution to raw images that matching training and inference conditions provides no additional benefit. CIRNet_FFA showed no improvement under matched conditions, remaining the worst performing retrained model across both raw and matched evaluations. CIRNet_USOD10K is included as the no enhancement reference point.

5.5.3 Effect of Enhancement at Inference

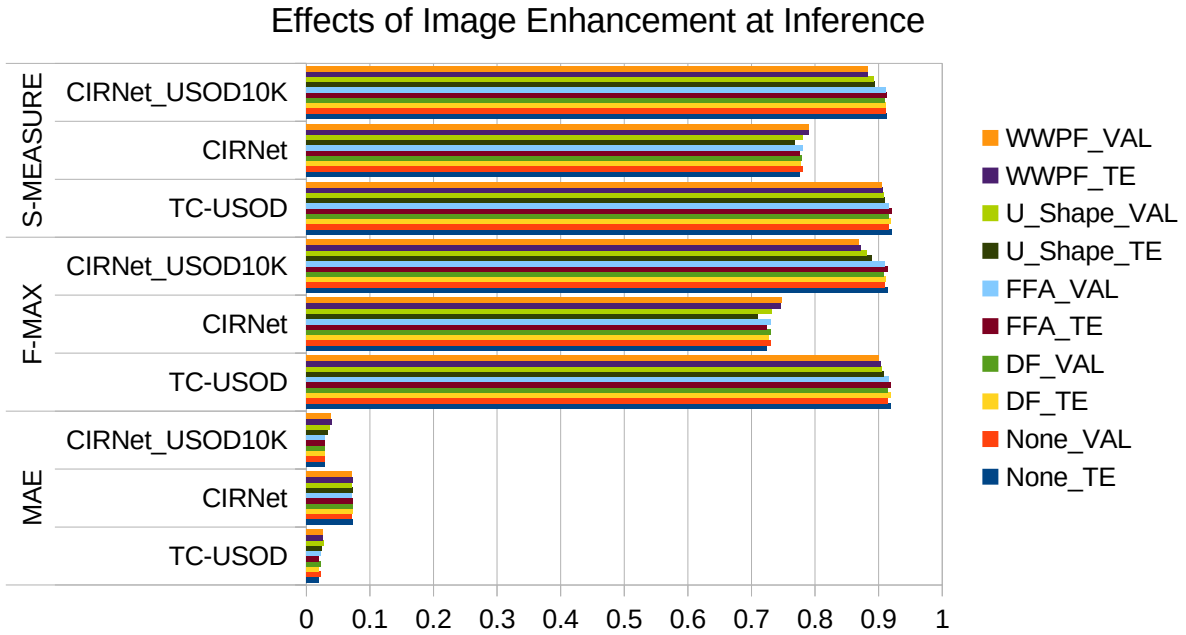


Figure 10: Pefromance of base models on enhanced images.

Figure 10 presents MAE, F-MAX and S-MEASURE for TC-USOD, CIRNet and CIRNet_USOD10K evaluated at inference on all five USOD10K variants. Full results across all metrics and both evaluation sets are provided in Appendix A.

TC-USOD demonstrated strong robustness to inference time enhancement, with DehazeFormer and FFA-Net producing effectively identical results to baseline across all three headline metrics. U-Shape Transformer and WWPF produced marginal degradation, with MAE increasing to 0.0240 and 0.0259 respectively.

CIRNet terrestrial weights showed minimal sensitivity to enhancement at inference, remaining consistently poor across all conditions with MAE ranging from 0.0718 to 0.0740, confirming that inference time enhancement alone cannot compensate for the lack of domain adaptation through retraining.

CIRNet_USOD10K showed a more interesting pattern , performance on DehazeFormer and FFA-Net enhanced images remained comparable to baseline, with MAE of 0.0291 and 0.0284 respectively. However U-Shape Transformer and WWPF enhanced images produced meaningful degradation, with MAE increasing to 0.0338 and 0.0395 respectively, mirroring the pattern observed in TC-USOD and consistent with the distribution shift argument. Both models were trained on unenhanced images and encounter a distribution mismatch when tested on strongly enhanced data.

6 Discussion and Conclusions

6.1 Perceptual Quality Metrics and SOD Performance

WWPF achieved the highest UCIQE and UIQM scores across all dataset splits, yet CIRNet models trained on WWPF enhanced images consistently underperformed those trained on DehazeFormer enhanced images on raw image evaluation. U-Shape Transformer scored second on both perceptual metrics yet produced comparable or worse SOD performance than FFA-Net across most conditions. This disconnect between perceptual quality improvement and downstream SOD performance is consistent with findings in the broader dehazing literature, where enhancement methods demonstrably improving image quality have been shown to have limited or negative effects on high level vision tasks [45].

The results suggest that UCIQE and UIQM, whilst valuable for evaluating human perceptual quality of underwater images, have no direct correlation with the features that SOD models depend on for accurate mask generation. What constitutes a visually appealing enhancement to human observers (increased colour saturation, improved brightness, vibrant colour restoration) does not necessarily preserve or enhance the edge contrast, boundary definition and spatial feature distributions that drive SOD performance. This raises the question of whether the underwater computer vision community would benefit from developing task specific image quality metrics that measure enhancement utility for downstream detection tasks rather than human perceptual quality alone.

6.2 Additive Nature of Domain Shifts

A consistent pattern emerged across U-Shape Transformer and WWPF enhanced conditions. Both methods produced strong performance improvements when training and inference images were matched, but degraded performance when applied only at inference to models trained on different distributions. CIRNet_U_Shape and CIRNet_WWPF were among the worst performing models when evaluated on raw or mismatched enhanced images, yet became competitive with CIRNet_USOD10K when evaluated on their own matched enhanced datasets, with CIRNet_WWPF achieving F-MAX of 0.9028 under matched conditions compared to 0.8499 on raw images.

This indicates that underwater specific enhancement methods introduce a secondary domain shift on top of the existing terrestrial to underwater domain gap. Rather than reducing the domain gap, these methods shift the image distribution to a new domain that the model must be specifically trained to expect. A model that has never seen U-Shape Transformer or WWPF enhanced images will not benefit from enhancement at inference and will in fact be penalised by it.

The practical implication is significant. Deploying an enhancement method alongside a SOD model in a real application requires that the SOD model was trained on images processed by that same enhancement method. Enhancement cannot be added as a post-hoc improvement to an already deployed model without retraining, at least for methods that produce strong distribution shifts. This partially negates the practical benefit of enhancement as a lightweight domain adaptation strategy, particularly for resource constrained deployment environments where running both an enhancement model and a SOD model simultaneously may be prohibitive.

The exception to this pattern is DehazeFormer, which is discussed in section 6.3.

6.3 Effects of Architecture on Generalisation

DehazeFormer represents the most surprising finding of this dissertation. Despite never being exposed to underwater imagery, CIRNet_DF achieved the closest performance to TC-USOD of all retrained variants on raw images, with F-MAX of 0.9154 compared to TC-USOD's 0.9177, a difference of less than 0.3%. Unlike U-Shape Transformer and WWPF, DehazeFormer enhanced images produced minimal distribution shift, with CIRNet_USOD10K showing comparable performance on DehazeFormer enhanced images to its performance on raw images at inference. This suggests that whatever DehazeFormer does to images is sufficiently subtle and consistent that models generalise across the transformation without requiring matched training conditions.

Visually, DehazeFormer's effect on underwater images can be characterised as a moderate contrast enhancement and partial haze reduction. It improves local feature visibility without dramatically altering the colour distribution or global image statistics. This is consistent with its transformer architecture learning generalised contrast and visibility enhancement features rather than atmospheric haze specific transformations.

The comparison with FFA-Net is particularly informative here. Both methods were trained on similar terrestrial haze datasets, yet their effects on downstream SOD performance diverge substantially. This connects to the broader argument established in section 4.4, transformers trained at scale tend to learn more transferable representations than CNNs trained on equivalent data, because their global attention mechanisms capture scene level statistical regularities rather than domain specific local features. DehazeFormer appears to have learned generalised visibility enhancement that transfers across degradation domains, while FFA-Net learned atmospheric haze removal specifically, producing transformations that are actively harmful to underwater SOD when the learned priors are misapplied.

This has an implication beyond this dissertation. Large scale terrestrial dehazing datasets may provide transferable enhancement capability to other degraded domains through transformer based models without requiring domain specific retraining, suggesting that data scarcity in underwater enhancement may be less limiting than previously assumed provided the right architecture is used.

6.4 Shortcomings of FFA-Net

FFA-Net produced the most consistently poor results of all enhancement methods, with CIRNet_FFA performing worse than CIRNet_USOD10K across all evaluation conditions including matched inference. This is particularly notable because FFA-Net showed minimal UCIQE improvement and only modest UIQM improvement over baseline, yet its effect on downstream SOD performance was substantially negative, suggesting it is not merely failing to improve underwater images but actively destroying information that CIRNet relies on for salient object detection.

Inspection of the prediction masks generated by CIRNet_FFA reveals a reduction in discriminability compared to other models, with shadow-like artefacts appearing around detected regions suggesting the model has learned uncertain or diffuse boundary representations from the FFA-Net processed training data, Figure 11. Examination of the FFA-Net processed RGB images themselves reveals a bimodal effect, some images appear comparable or marginally improved, while others exhibit

severe chromatic distortion, with unnatural hue shifts producing visually aberrant colour casts across the image. This inconsistency in enhancement output likely contributed to inconsistent and degraded training signal, producing a model that learned neither clean underwater features nor consistently enhanced ones.

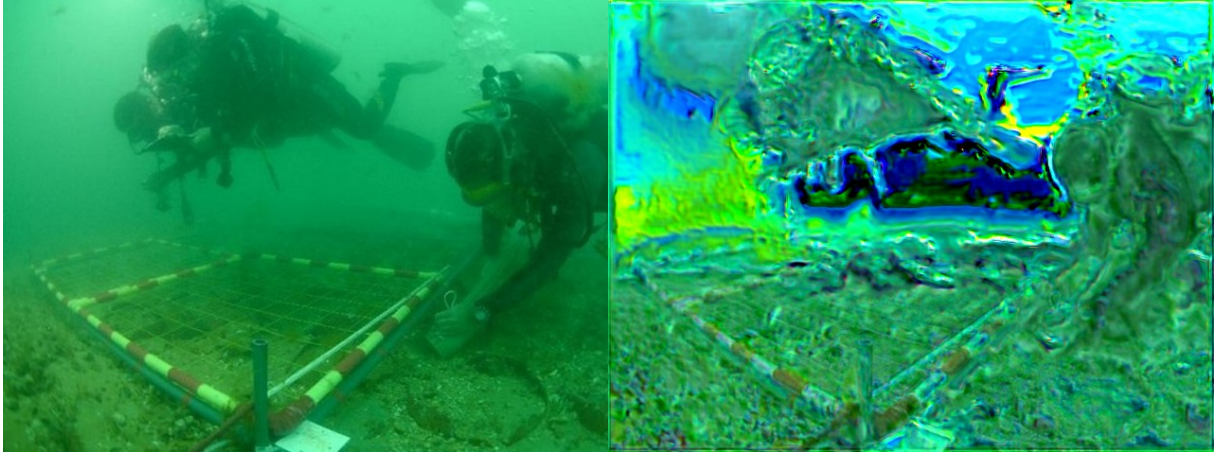


Figure 11: Example image of FFA-Net failure. Left: Base USOD10K image. Right: FFA-Net enhanced image.

The contrast with DehazeFormer is striking given both methods were trained on similar terrestrial haze datasets. Where DehazeFormer produced consistent moderate enhancement that transferred beneficially to underwater SOD, FFA-Net produced inconsistent and sometimes destructive transformations. This further supports the architectural generalisation argument from section 6.3, the issue is not the training data but what each architecture learned from it.

6.5 Architecture Remains the Primary Performance Determinant

Across all experimental conditions TC-USOD maintained a consistent performance advantage over all CIRNet variants on MAE and the primary SOD metrics, confirming that architectural specialisation for the underwater domain provides benefits that preprocessing alone cannot fully replicate. Even CIRNet_DF, the best performing retrained variant, showed a 21% relative degradation in MAE compared to TC-USOD, the largest proportional gap of any primary metric, suggesting that whilst F-MAX and S-MEASURE differences are minor, error rates in absolute mask accuracy remain meaningfully higher for the CNN based approach.

However the performance gap is substantially smaller than the USOD10K paper's benchmark results for terrestrial models retrained on USOD10K would suggest. CIRNet retrained on raw USOD10K images achieved F-MAX of 0.9123 and S-MEASURE of 0.9123, which is competitive with TC-USOD's 0.9177 and 0.9183 respectively. This raises the question of whether the models used as competitors in the original benchmark were representative of current state of the art CNN based SOD, or whether the field has advanced sufficiently since those comparisons were made that a more recent CNN based model with attention mechanisms narrows the gap considerably.

This finding supports the hypothesis established in section 4.4. At the data scales available in underwater SOD, the theoretical transformer advantage is largely unrealised on most metrics. CIRNet's incorporation of attention mechanisms that approximate transformer self attention behaviour may be a significant factor here, suggesting that the architectural boundary between CNN and transformer based approaches is less sharp in practice than the categorical distinction implies. The question may not be CNN versus transformer but rather how much attention capacity is necessary and at what computational cost.

Enhancement can provide marginal improvements to an already capable model but cannot compensate for fundamental architectural limitations. The conclusion that architecture is the primary performance determinant is consistent with the broader SOD literature, but the finding that a terrestrial CNN retrained on underwater data can approach specialised transformer performance on most metrics at this dataset scale is a meaningful result for the field, particularly given the practical deployment advantages of CNN based approaches discussed in section 4.4.

6.6 Practical and Deployment Considerations

Several observations from the experimental results have direct implications for practical deployment of USOD systems that extend beyond benchmark performance comparisons

TC-USOD inference time on enhanced images was consistently approximately 20% faster than on raw images, averaging approximately 16ms per image on processed datasets compared to 21ms on raw images. This reduction was consistent across all enhancement methods regardless of the degree of distribution shift each method introduced. A plausible explanation is that TC-USOD performs internal preprocessing as part of its architecture to normalise underwater degradation, and that pre-enhanced images require less internal correction, effectively allowing the model to skip or reduce that processing stage. The consistency of this reduction across enhancement methods suggests it reflects a general property of the model's internal representation rather than a specific interaction with any particular enhancement approach. This is an unexpected finding that warrants further investigation.

The resolution of images used throughout this experiment was 480x640 or 640x480 pixels, which is substantially below the resolution of modern imaging hardware. The computational complexity difference between CNN and transformer based approaches, $O(n)$ versus $O(n^2)$ respectively, would become increasingly significant at higher resolutions representative of real deployment conditions. Benchmark comparisons at low resolution may therefore understate the practical computational advantage of CNN based approaches for real time deployment, and the performance gap between architectures may manifest differently at resolutions more representative of actual underwater imaging systems.

USOD10K itself exhibits notable inconsistency in effective image quality despite uniform pixel dimensions, with some images appearing to be crops expanded from higher resolution sources and others compressed from larger originals. This introduces uncontrolled variance in the training and evaluation data that may affect model performance in ways that are difficult to isolate. A controlled experiment varying effective image resolution whilst holding pixel dimensions constant would help clarify whether this is a meaningful factor.

AUC and AP showed consistently high scores with minimal variation across all models and conditions, suggesting these metrics provide limited discriminative power for USOD evaluation specifically. Both metrics measure ranking ability across thresholds rather than absolute mask accuracy, and the consistently high scores suggest all models regardless of quality produce well ranked saliency predictions even when absolute mask accuracy varies substantially. MAE, F-MAX and S-MEASURE are recommended as the primary evaluation metrics for future USOD work.

The requirement for matched training and inference enhancement conditions identified in section 6.2 has a direct deployment implication. Any system using enhancement as a preprocessing strategy must run both the enhancement model and the SOD model at inference time, increasing computational overhead and deployment complexity. For real time applications this may negate the performance benefit entirely. For slower processing pipelines such as batch analysis of recorded footage the improvement in performance may justify the additional overhead, particularly given WWPF's strong matched condition results and its parameter free nature which eliminates the need for a second neural network at inference.

6.7 Future Work

Several directions for future investigation emerge directly from the findings of this dissertation.

The absence of MVG-FD from the experimental pipeline due to unavailable pretrained weights represents the most immediate gap. MVG-FD is a direct development on U-Shape Transformer incorporating multimodal depth guidance, and its inclusion would allow direct comparison between successive generations of underwater specific enhancement methods, clarifying whether architectural advances in enhancement translate to meaningful downstream SOD improvements. Retraining MVG-FD on LSUI and evaluating it within the same experimental framework would complete the enhancement spectrum originally intended.

The finding that FFA-Net retrained on LSUI was not completed due to time constraints leaves an important question unanswered. A supervised enhancement model retrained specifically on underwater paired data would provide a direct comparison between terrestrial pretrained and domain matched supervised enhancement, isolating the contribution of training domain from architectural differences. This comparison would clarify whether DehazeFormer's advantage over FFA-Net reflects architectural generalisation capacity or simply the consequence of FFA-Net never seeing underwater data in any form.

The resolution experiment suggested by USOD10K's inconsistent effective image quality warrants dedicated investigation. Testing the same models on controlled upscaled and downscaled versions of the same images whilst holding pixel dimensions constant would isolate the effect of effective image resolution on SOD performance, and would provide empirical data on how the $O(n)$ versus $O(n^2)$ complexity difference between CNN and transformer architectures manifests at resolutions representative of modern imaging hardware.

The inference speed finding for TC-USOD warrants further investigation. The consistent 20% reduction in inference time across all enhanced image conditions suggests an interaction between preprocessing and TC-USOD's internal architecture that is not fully understood. A systematic

analysis of TC-USOD's internal processing pipeline on raw versus enhanced inputs would clarify whether this reflects genuine computational savings or a measurement artefact.

The disconnect between UCIQE and UIQM scores and downstream SOD performance identified in section 6.1 motivates the development of task specific image quality metrics for underwater computer vision. Such metrics would measure enhancement utility for detection tasks directly rather than proxying through human perceptual quality, providing more meaningful guidance for enhancement method selection in computer vision pipelines. Investigation of which specific components of UCIQE and UIQM, if any, correlate with SOD performance improvements would be a productive starting point.

Finally the application specific annotation point raised in section 4 remains an open challenge. USOD10K provides general saliency annotations but real deployment scenarios including marine infrastructure inspection, search and rescue, and ecological monitoring require purpose specific ground truth that may differ substantially from general saliency judgements. Development of application specific USOD subsets from the existing USOD10K data, or collection of purpose annotated datasets for specific underwater applications, would enable more targeted model development and evaluation.

6.8 Conclusion

This dissertation investigated whether image enhancement can bridge the domain gap between terrestrial SOD models and underwater scenes, and whether CNN based models remain viable for USOD under realistic data constraints.

The results demonstrate that image enhancement can provide meaningful improvements to USOD performance, but only under specific conditions. Enhancement methods that introduce strong distribution shifts, specifically U-Shape Transformer and WWPF, improve performance substantially when training and inference conditions are matched but actively degrade performance otherwise. DehazeFormer, despite never being exposed to underwater imagery, produced the most consistent improvements across conditions, suggesting that transformer based enhancement methods trained at scale develop transferable generalisation capacity that CNN based equivalents do not.

The hypothesis that the theoretical transformer advantage is unrealised at underwater dataset scales was supported. CIRNet retrained on USOD10K achieved performance within 1% of TC-USOD on F-MAX and S-MEASURE under matched conditions, suggesting that at 10k images architectural specialisation provides marginal rather than decisive advantage on most metrics. MAE remains the exception, where a meaningful gap persists.

The finding that perceptual image quality metrics have no reliable correlation with downstream SOD performance is a practically significant result. UCIQE and UIQM scores did not predict which enhancement methods would improve detection performance, and in some cases the highest scoring methods produced the worst SOD results. This has direct implications for how enhancement methods are selected and evaluated in underwater computer vision pipelines.

In conclusion, architectural specialisation remains the primary determinant of USOD performance, but a capable CNN based model retrained on underwater data can approach specialised transformer

performance on most metrics at current dataset scales. Enhancement can provide marginal additional improvements but cannot substitute for architectural adaptation, and the choice of enhancement method matters as much as its presence. As underwater datasets grow in scale and variety, the relative advantage of transformer based approaches will likely increase, making continued investment in specialised underwater architectures worthwhile for the long term whilst CNN based approaches with preprocessing remain a practical and competitive option for near term deployment.

7 Project Review

7.1 What Went Well

The core experimental pipeline was completed within the available time and produced a full set of results across all planned model and enhancement method combinations. The decision to use pretrained enhancement models rather than training new ones proved correct; it kept the preprocessing pipeline manageable and focused the computational budget on the SOD training runs where the primary research questions were located.

The use of Docker containers for environment management was effective, preventing dependency conflicts between models with incompatible requirements and ensuring that each model ran in a consistent and reproducible environment.

USOD10K was selected as the training and evaluation dataset, consistent with the project specification which identified it as a suitable benchmark. This proved to be a strong foundation for the experiment; the dataset's scale, the simultaneous release of TC-USOD as a specialised baseline, and its rapid adoption as the field standard made it well suited to the research questions investigated.

7.2 Setbacks

UC-Net [51] represented the most significant setback. Three separate training attempts failed with NaN tensor outputs at 8, 15 and 23 epochs respectively. Each failure was initially attributed to a different cause: GPU selection configuration at 8 epochs, learning rate instability at 15 epochs, and an unidentified deeper issue at 23 epochs. In retrospect the repeated failures suggested an underlying compatibility issue rather than configuration errors. Post-hoc analysis suggests that insufficient attention was paid to ROCm version compatibility with the available hardware, and that the use of a Python 2 dependent codebase, a version long since superseded, should have been treated as an immediate red flag. Had these compatibility concerns been identified earlier the decision to discard UC-Net could have been made after the first failure rather than the third, saving significant time.

DCP was expected to perform poorly given the known incompatibility between atmospheric scattering priors and underwater imagery, however the severity of the failure was not anticipated. Rather than producing degraded or low quality enhancement, DCP destroyed image content entirely. Large portions of images were replaced with solid blocks of colour, rendering them completely unusable. This suggests that the underwater colour channel statistics so severely violate

DCP's core prior assumptions that the transmission map estimation collapses entirely rather than simply producing inaccurate results. Testing on a small sample before committing to full dataset processing would have identified this earlier and saved the time spent processing the complete dataset.

MVG-FD [48] was identified as a potentially valuable addition but had no pretrained weights available, meaning it could not be included without training from scratch which fell outside the available time and scope.

The hardware constraint of a single consumer GPU meant total training time of approximately 219 hours, which limited the number of experimental conditions that could be explored. Specifically FFA-Net retraining on LSUI, which would have provided a domain matched supervised baseline, was considered but could not be completed within the available time.

7.3 Project Improvements

With access to more computational resources the experiment could be extended in several directions. Retraining FFA-Net or a comparable supervised method on LSUI would add a domain matched supervised enhancement baseline, allowing direct comparison between terrestrial pretrained and underwater trained enhancement methods. Training MVG-FD from scratch would provide the cutting edge multimodal baseline originally intended.

The literature search process could be made significantly more efficient. Many recent papers do not list associated repositories in the paper itself. Searching GitHub directly using model names and keywords earlier in the process would have identified available codebases faster. Additionally, searching for image enhancement rather than dehazing would have surfaced relevant papers that never use the term dehazing despite addressing the same underlying problem, as terminology varies across research groups.

A more systematic approach to model compatibility should be adopted from the outset. Any model with ROCm or CUDA version dependencies should be tested for hardware compatibility before investing training time. Similarly any codebase relying on deprecated Python versions should be treated as high risk and either replaced or thoroughly audited before use. A short pilot training run of no more than 10 epochs on any new model before committing to full training would identify instability issues early and avoid the repeated failure pattern encountered with UC-Net.

8 Acknowledgements

I would like to thank Dr Adrian Evans for his guidance and support throughout this project. I would also like to thank my parents for continuing to push me to achieve more than I deem possible. I would also like to thank my dog, for helping me keep my sanity.

8.1 Use of Generative Artificial Intelligence

Claude (Anthropic) was used during the preparation of this dissertation to assist with structuring arguments, drafting and refining written sections, and discussing technical concepts. All

experimental work, data collection, analysis, and final intellectual decisions remain my own. All AI assisted content was reviewed, verified and edited by me prior to inclusion.

9 List of References

- [1] R. Cong *et al.*, ‘CIR-Net: Cross-Modality Interaction and Refinement for RGB-D Salient Object Detection’, *IEEE Transactions on Image Processing*, vol. 31, pp. 6800–6815, 2022, doi: 10.1109/TIP.2022.3216198.
- [2] R. Ju, L. Ge, W. Geng, T. Ren, and G. Wu, ‘Depth saliency based on anisotropic center-surround difference’, in *2014 IEEE International Conference on Image Processing (ICIP)*, Oct. 2014, pp. 1115–1119. doi: 10.1109/ICIP.2014.7025222.
- [3] L. Hong, X. Wang, G. Zhang, and M. Zhao, ‘USOD10K: A New Benchmark Dataset for Underwater Salient Object Detection’, *IEEE Transactions on Image Processing*, vol. 34, pp. 1602–1615, 2025, doi: 10.1109/TIP.2023.3266163.
- [4] L. Peng, C. Zhu, and L. Bian, ‘U-shape Transformer for Underwater Image Enhancement’, *IEEE Trans. on Image Process.*, vol. 32, pp. 3066–3079, 2023, doi: 10.1109/TIP.2023.3276332.
- [5] J. Y. Chiang and Y.-C. Chen, ‘Underwater Image Enhancement by Wavelength Compensation and Dehazing’, *IEEE transactions on image processing : a publication of the IEEE Signal Processing Society*, vol. 21, pp. 1756–69, Dec. 2011, doi: 10.1109/TIP.2011.2179666.
- [6] A. Borji, ‘What is a Salient Object? A Dataset and a Baseline Model for Salient Object Detection’, *IEEE Transactions on Image Processing*, vol. 24, no. 2, pp. 742–756, Feb. 2015, doi: 10.1109/TIP.2014.2383320.
- [7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, ‘You Only Look Once: Unified, Real-Time Object Detection’, May 09, 2016, *arXiv*: arXiv:1506.02640. doi: 10.48550/arXiv.1506.02640.
- [8] H. Fang *et al.*, ‘From Captions to Visual Concepts and Back’, Apr. 14, 2015, *arXiv*: arXiv:1411.4952. doi: 10.48550/arXiv.1411.4952.
- [9] A. Borji, S. Frntrop, D. N. Sihite, and L. Itti, ‘Adaptive object tracking by learning background context’, in *2012 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*, Jun. 2012, pp. 23–30. doi: 10.1109/CVPRW.2012.6239191.
- [10] Y. Lin, Z. Pang, D. Wang, and Y. Zhuang, ‘Task-driven Visual Saliency and Attention-based Visual Question Answering’, Feb. 22, 2017, *arXiv*: arXiv:1702.06700. doi: 10.48550/arXiv.1702.06700.
- [11] R. Zhao, W. Oyang, and X. Wang, ‘Person Re-Identification by Saliency Learning’, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 2, pp. 356–370, Feb. 2017, doi: 10.1109/TPAMI.2016.2544310.
- [12] X. Yang, H. Kong, and Y. Meng, ‘Salient Object Detection in Traffic Scenes Based on RGB-D Fusion’, in *2024 8th CAA International Conference on Vehicular Control and Intelligence (CVCI)*, Oct. 2024, pp. 1–5. doi: 10.1109/CVCI63518.2024.10830049.
- [13] L. Itti, C. Koch, and E. Niebur, ‘A model of saliency-based visual attention for rapid scene analysis’, *IEEE Trans. Pattern Anal. Machine Intell.*, vol. 20, no. 11, pp. 1254–1259, Nov. 1998, doi: 10.1109/34.730558.
- [14] J. Xu, M. Jiang, S. Wang, M. S. Kankanhalli, and Q. Zhao, ‘Predicting human gaze beyond pixels’, *Journal of Vision*, vol. 14, no. 1, p. 28, Jan. 2014, doi: 10.1167/14.1.28.
- [15] F. Zhang *et al.*, ‘Revisiting Salient Object Detection from an Observer-Centric Perspective’, Feb. 06, 2026, *arXiv*: arXiv:2602.06369. doi: 10.48550/arXiv.2602.06369.
- [16] A. Chen, X. Li, T. He, J. Zhou, and D. Chen, ‘Advancing in RGB-D Salient Object Detection: A Survey’, *Applied Sciences*, vol. 14, no. 17, p. 8078, Jan. 2024, doi: 10.3390/app14178078.

- [17] G. Li and Y. Yu, ‘Visual Saliency Detection Based on Multiscale Deep CNN Features’, *IEEE Trans. on Image Process.*, vol. 25, no. 11, pp. 5012–5024, Nov. 2016, doi: 10.1109/TIP.2016.2602079.
- [18] L. Wang, H. Lu, X. Ruan, and M.-H. Yang, ‘Deep networks for saliency detection via local estimation and global search’, in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2015, pp. 3183–3192. doi: 10.1109/CVPR.2015.7298938.
- [19] R. Zhao, W. Ouyang, H. Li, and X. Wang, ‘Saliency detection by multi-context deep learning’, in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2015, pp. 1265–1274. doi: 10.1109/CVPR.2015.7298731.
- [20] C. Sun, A. Shrivastava, S. Singh, and A. Gupta, ‘Revisiting Unreasonable Effectiveness of Data in Deep Learning Era’, in *2017 IEEE International Conference on Computer Vision (ICCV)*, Venice: IEEE, Oct. 2017, pp. 843–852. doi: 10.1109/ICCV.2017.97.
- [21] A. Dosovitskiy *et al.*, ‘An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale’, Jun. 03, 2021, *arXiv*: arXiv:2010.11929. doi: 10.48550/arXiv.2010.11929.
- [22] taozh2017, ‘RGBD-SODsurvey/README.md at master · taozh2017/RGBD-SODsurvey’, GitHub. Accessed: Apr. 07, 2026. [Online]. Available: <https://github.com/taozh2017/RGBD-SODsurvey/blob/master/README.md>
- [23] X. Li, J. Hao, M. Shang, and Z. Yang, ‘Saliency segmentation and foreground extraction of underwater image based on localization’, in *OCEANS 2016 - Shanghai*, Apr. 2016, pp. 1–4. doi: 10.1109/OCEANSAP.2016.7485498.
- [24] Z. Chen, Y. Sun, Y. Gu, H. Wang, H. Qian, and H. Zheng, ‘Underwater Object Segmentation Integrating Transmission and Saliency Features’, *IEEE Access*, vol. 7, pp. 72420–72430, 2019, doi: 10.1109/ACCESS.2019.2919711.
- [25] R. Chen, Z. Fu, Y. Huang, E. Cheng, and X. Ding, ‘A Robust Object Segmentation Network for UnderWater Scenes’, in *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022, pp. 2629–2633. doi: 10.1109/ICASSP43922.2022.9746176.
- [26] H. Feng, L. Xu, X. Yin, and Z. Chen, ‘Underwater salient object detection based on red channel correction’, in *2021 IEEE 2nd International Conference on Big Data, Artificial Intelligence and Internet of Things Engineering (ICBAIE)*, Mar. 2021, pp. 446–449. doi: 10.1109/ICBAIE52039.2021.9390003.
- [27] Q. Wu, J. Xie, Z. Fu, X. Tu, Y. Huang, and X. Ding, ‘CE³USOD: Channel-Enhanced, Efficient, and Effective Network for Underwater Salient Object Detection’, *IEEE Journal of Oceanic Engineering*, vol. 50, no. 2, pp. 941–954, Apr. 2025, doi: 10.1109/JOE.2024.3523356.
- [28] L. Lu, X. Xiong, L. Liu, and H. Tang, ‘BG-UISNet: A Novel Boundary-Guided Network for Robust and Precise Underwater Image Segmentation’, *IEEE Sensors Journal*, vol. 25, no. 17, pp. 33387–33396, Sep. 2025, doi: 10.1109/JSEN.2025.3593256.
- [29] Y. Liu, X. Zhang, J. Zhu, B. Ma, Y. Duan, and P. Tan, ‘HDANet: Enhancing Underwater Salient Object Detection With Physics-Inspired Multimodal Joint Learning’, *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–14, 2025, doi: 10.1109/TGRS.2025.3565579.
- [30] Z. Mi *et al.*, ‘Interaction-Driven Edge Crisping for Underwater Salient Object Detection’, *IEEE Transactions on Circuits and Systems for Video Technology*, pp. 1–1, 2026, doi: 10.1109/TCSVT.2026.3679543.
- [31] Y. Ganin *et al.*, ‘Domain-Adversarial Training of Neural Networks’, May 26, 2016, *arXiv*: arXiv:1505.07818. doi: 10.48550/arXiv.1505.07818.
- [32] S. Salehnia, O. Tastan, and E. Aptoula, ‘Frequency Domain Image Augmentation for Domain Generalized Image Classification’, in *2025 33rd Signal Processing and Communications Applications Conference (SIU)*, Jun. 2025, pp. 1–4. doi: 10.1109/SIU66497.2025.11112033.

- [33] Y. Luo, J. Feinglass, T. Gokhale, K.-C. Lee, C. Baral, and Y. Yang, ‘Grounding Stylistic Domain Generalization with Quantitative Domain Shift Measures and Synthetic Scene Images’, in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Jun. 2024, pp. 7303–7313. doi: 10.1109/CVPRW63382.2024.00726.
- [34] T. Razzaq and A. Iqbal, ‘Multimodal Neuromorphic Event-Frame Fusion in Domain-Generalized Vision Transformer for Dynamic Object Tracking’, in *2025 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, Oct. 2025, pp. 4743–4750. doi: 10.1109/ICCVW69036.2025.00492.
- [35] S. Salehnia, O. Tastan, and E. Aptoula, ‘APA: Domain Generalization Using Frequency Based Augmentation’, in *2025 IEEE 35th International Workshop on Machine Learning for Signal Processing (MLSP)*, Aug. 2025, pp. 1–6. doi: 10.1109/MLSP62443.2025.11204305.
- [36] P. Vaish, S. Wang, and N. Strisciuglio, ‘Fourier-Basis Functions to Bridge Augmentation Gap: Rethinking Frequency Augmentation in Image Classification’, in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2024, pp. 17763–17772. doi: 10.1109/CVPR52733.2024.01682.
- [37] N. O. and A. A. US Department of Commerce, ‘What is marine snow?’ Accessed: May 03, 2026. [Online]. Available: <https://oceanservice.noaa.gov/facts/marinesnow.html>
- [38] A. Krizhevsky, I. Sutskever, and G. E. Hinton, ‘ImageNet classification with deep convolutional neural networks’, *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017, doi: 10.1145/3065386.
- [39] T.-Y. Lin *et al.*, ‘Microsoft COCO: Common Objects in Context’, Feb. 21, 2015, *arXiv*: arXiv:1405.0312. doi: 10.48550/arXiv.1405.0312.
- [40] A. Kuznetsova *et al.*, ‘The Open Images Dataset V4: Unified image classification, object detection, and visual relationship detection at scale’, *Int J Comput Vis*, vol. 128, no. 7, pp. 1956–1981, Jul. 2020, doi: 10.1007/s11263-020-01316-z.
- [41] A. Vaswani *et al.*, ‘Attention Is All You Need’, Aug. 02, 2023, *arXiv*: arXiv:1706.03762. doi: 10.48550/arXiv.1706.03762.
- [42] I. Aguilera-Martos, A. Herrera-Poyatos, J. Luengo, and F. Herrera, ‘Local Attention Mechanism: Boosting the Transformer Architecture for Long-Sequence Time Series Forecasting’, 2024, *arXiv*. doi: 10.48550/ARXIV.2410.03805.
- [43] K. He, J. Sun, and X. Tang, ‘Single Image Haze Removal Using Dark Channel Prior’, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011, doi: 10.1109/TPAMI.2010.168.
- [44] W. Zhang *et al.*, ‘Underwater Image Enhancement via Weighted Wavelet Visual Perception Fusion’, *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 4, pp. 2469–2483, Apr. 2024, doi: 10.1109/TCSVT.2023.3299314.
- [45] J. Gui *et al.*, ‘A Comprehensive Survey and Taxonomy on Single Image Dehazing Based on Deep Learning’, *ACM Computing Surveys*, vol. 55, no. 13s, pp. 1–37, Jul. 2023, doi: 10.1145/3576918.
- [46] M. Yang and A. Sowmya, ‘An Underwater Color Image Quality Evaluation Metric’, *IEEE Transactions on Image Processing*, vol. 24, no. 12, pp. 6062–6071, Dec. 2015, doi: 10.1109/TIP.2015.2491020.
- [47] K. Panetta, C. Gao, and S. Agaian, ‘Human-Visual-System-Inspired Underwater Image Quality Measures’, *IEEE Journal of Oceanic Engineering*, vol. 41, no. 3, pp. 541–551, Jul. 2016, doi: 10.1109/JOE.2015.2469915.
- [48] G. Jiang, A. Zhang, Y. Liu, H. Wang, and S. Xu, ‘MVG-FD: Multi-Modal Visual Guidance and Feature Decomposition for Underwater Image Restoration’, *IEEE Signal Processing Letters*, vol. 32, pp. 3305–3309, 2025, doi: 10.1109/LSP.2025.3596447.

- [49] Y. Song, Z. He, H. Qian, and X. Du, ‘Vision Transformers for Single Image Dehazing’, *IEEE Transactions on Image Processing*, vol. 32, pp. 1927–1941, 2023, doi: 10.1109/TIP.2023.3256763.
- [50] X. Qin, Z. Wang, Y. Bai, X. Xie, and H. Jia, ‘FFA-Net: Feature Fusion Attention Network for Single Image Dehazing’, Dec. 05, 2019, *arXiv*: arXiv:1911.07559. doi: 10.48550/arXiv.1911.07559.
- [51] J. Zhang *et al.*, ‘Uncertainty Inspired RGB-D Saliency Detection’, Sep. 07, 2020, *arXiv*: arXiv:2009.03075. doi: 10.48550/arXiv.2009.03075.

10 Appendices

10.1 Appendix A – Full Results Tables

MAE	Model/ Enhancment	CIRNet_USOD				CIRNet_U		
		TC-USOD	CIRNet	10K	CIRNet_DF	CIRNet_FFA	_Shape	CIRNet_WWPF
F-MAX	None_TE	0.0203	0.0725	0.0284	0.0263	0.0413	0.0417	0.0423
	None_VAL	0.0235	0.0721	0.0290	0.0270	0.0406	0.0418	0.0418
	DF_TE	0.0205	0.0726	0.0291	0.0271	0.0412	0.045	0.042
	DF_VAL	0.0235	0.0728	0.0293	0.0273	0.0405	0.0433	0.0417
	FFA_TE	0.0203	0.0725	0.0284	0.0263	0.0413	0.0417	0.0423
	FFA_VAL	0.0234	0.0721	0.0290	0.0270	0.0406	0.0418	0.0418
	U_Shape_TE	0.024	0.074	0.0338	0.0322	0.0467	0.0353	0.042
	U_Shape_VAL	0.0271	0.0718	0.0367	0.0336	0.0483	0.0338	0.0402
	WWPF_TE	0.0259	0.0729	0.0395	0.0369	0.0487	0.0639	0.0316
	WWPF_VAL	0.0267	0.0724	0.0390	0.0365	0.0492	0.0609	0.0308
F-MEAN	None_TE	0.9203	0.7242	0.9145	0.9167	0.8842	0.8763	0.8497
	None_VAL	0.915	0.7312	0.9101	0.9141	0.8873	0.8718	0.85
	DF_TE	0.9192	0.7281	0.9115	0.9116	0.8833	0.8635	0.8475
	DF_VAL	0.915	0.731	0.9080	0.9117	0.888	0.8673	0.8488
	FFA_TE	0.9203	0.7242	0.9145	0.9167	0.8842	0.8763	0.8497
	FFA_VAL	0.9155	0.7312	0.9101	0.9141	0.8873	0.8718	0.85
	U_Shape_TE	0.9089	0.7098	0.8894	0.8924	0.857	0.8901	0.8589
	U_Shape_VAL	0.9047	0.7327	0.8825	0.8876	0.8609	0.8948	0.8582
	WWPF_TE	0.9035	0.7459	0.8719	0.8817	0.858	0.7878	0.9018
	WWPF_VAL	0.9014	0.7476	0.8693	0.8785	0.8562	0.7907	0.9038
E-MAX	None_TE	0.9044	0.6877	0.8831	0.8891	0.8376	0.8394	0.8171
	None_VAL	0.8988	0.6947	0.8781	0.8853	0.84	0.8359	0.8173
	DF_TE	0.9033	0.6919	0.8807	0.8848	0.8369	0.8283	0.8158
	DF_VAL	0.8988	0.6948	0.8757	0.8829	0.8403	0.83	0.8174
	FFA_TE	0.9044	0.6877	0.8831	0.8891	0.8376	0.8394	0.8171
	FFA_VAL	0.8994	0.6947	0.8781	0.8853	0.84	0.8359	0.8173
	U_Shape_TE	0.8883	0.6745	0.8589	0.8648	0.8108	0.8546	0.8266
	U_Shape_VAL	0.8836	0.6963	0.8514	0.8598	0.8155	0.8573	0.8259
	WWPF_TE	0.8841	0.7071	0.8388	0.8506	0.8046	0.752	0.8668
	WWPF_VAL	0.8824	0.7093	0.8374	0.8469	0.8057	0.7542	0.8683

	DF_VAL	0.9601	0.8394	0.9527	0.9545	0.9397	0.9246	0.9122
	FFA_TE	0.9653	0.8341	0.9579	0.9602	0.9387	0.9308	0.912
	FFA_VAL	0.9604	0.8399	0.9543	0.9562	0.9389	0.928	0.9127
	U_Shape_TE	0.9573	0.8297	0.9432	0.9418	0.9242	0.9427	0.9203
	U_Shape_VAL	0.9537	0.8433	0.9369	0.9401	0.9224	0.947	0.9215
	WWPF_TE	0.9556	0.8467	0.9283	0.9372	0.9217	0.8682	0.9512
	WWPF_VAL	0.9516	0.8465	0.9284	0.9338	0.9222	0.8743	0.9515
	E-MEAN							
	None_TE	0.96	0.7836	0.9381	0.9419	0.9027	0.9021	0.8807
	None_VAL	0.9547	0.7885	0.9349	0.9388	0.9038	0.8995	0.8801
	DF_TE	0.9592	0.7861	0.9353	0.9384	0.9026	0.894	0.8793
	DF_VAL	0.9547	0.7885	0.9330	0.9371	0.9046	0.895	0.8794
	FFA_TE	0.96	0.7836	0.9381	0.9419	0.9027	0.9021	0.8807
	FFA_VAL	0.9551	0.7885	0.9349	0.9388	0.9038	0.8995	0.8801
	U_Shape_TE	0.9511	0.7809	0.9219	0.9244	0.8863	0.9169	0.8886
	U_Shape_VAL	0.9473	0.7907	0.9167	0.9222	0.8875	0.9199	0.8915
	WWPF_TE	0.9499	0.8018	0.9075	0.9160	0.881	0.8365	0.9278
	WWPF_VAL	0.9461	0.8027	0.9072	0.9135	0.8824	0.8427	0.9283
	S-MEASURE							
	RE							
	None_TE	0.9204	0.7768	0.9132	0.9163	0.8909	0.886	0.8697
	None_VAL	0.9161	0.7807	0.9114	0.9141	0.8918	0.8855	0.871
	DF_TE	0.92	0.7778	0.9112	0.9131	0.891	0.8796	0.8689
	DF_VAL	0.9161	0.7801	0.9097	0.9126	0.892	0.8812	0.8707
	FFA_TE	0.9204	0.7768	0.9132	0.9163	0.8909	0.886	0.8697
	FFA_VAL	0.917	0.7807	0.9114	0.9141	0.8918	0.8855	0.871
	U_Shape_TE	0.9098	0.7689	0.8946	0.8973	0.8724	0.8974	0.8739
	U_Shape_VAL	0.908	0.7816	0.8922	0.8965	0.8752	0.9007	0.8775
	AP							
	WWPF_TE	0.9061	0.79	0.8828	0.8882	0.8707	0.8274	0.9066
	WWPF_VAL	0.9046	0.7905	0.8835	0.8884	0.8716	0.8324	0.9077
	AUC							
	None_TE	0.9123	0.7407	0.9519	0.9529	0.9385	0.9196	0.8906
	None_VAL	0.9095	0.7378	0.9478	0.9493	0.942	0.9195	0.8862
	DF_TE	0.91	0.7474	0.9493	0.9497	0.9379	0.9137	0.8907
	DF_VAL	0.9095	0.74	0.9443	0.9476	0.9433	0.9114	0.8847
	FFA_TE	0.9123	0.7407	0.9519	0.9529	0.9385	0.9196	0.8906
	FFA_VAL	0.9106	0.7378	0.9478	0.9493	0.942	0.9195	0.8862
	U_Shape_TE	0.9016	0.7195	0.9226	0.9249	0.9159	0.9355	0.9011
	U_Shape_VAL	0.9005	0.7421	0.9185	0.9254	0.9224	0.9394	0.8959
	WWPF_TE	0.8918	0.7761	0.9122	0.9202	0.9195	0.8274	0.9441
	WWPF_VAL	0.8913	0.7742	0.9095	0.9183	0.9166	0.8305	0.9444
	AUC							
	None_TE	0.9694	0.9177	0.9864	0.9859	0.9878	0.9787	0.9677
	None_VAL	0.9677	0.9146	0.9862	0.9852	0.9875	0.9792	0.9657
	DF_TE	0.9689	0.9195	0.9851	0.9849	0.9881	0.9767	0.9678
	DF_VAL	0.9677	0.9152	0.9853	0.9848	0.9881	0.9766	0.9655
	FFA_TE	0.9694	0.9177	0.9864	0.9859	0.9878	0.9787	0.9677
	FFA_VAL	0.9683	0.9146	0.9862	0.9852	0.9875	0.9792	0.9657
	U_Shape_TE	0.965	0.9119	0.9763	0.9764	0.9814	0.9827	0.9691

	U_Shape_VAL	0.9654	0.9145	0.9772	0.9786	0.9821	0.9856	0.9715
	WWPF_TE	0.9627	0.9279	0.9753	0.9760	0.9841	0.95	0.986
	WWPF_VAL	0.9631	0.9274	0.9751	0.9759	0.9838	0.951	0.9865
UCIQE	Set	TE	VAL	TR	Average			
	None	0.5324	0.5335	0.5336	0.5332			
	DF	0.5422	0.5438	0.5436	0.5432			
	FFA	0.5324	0.5335	0.5479	0.5379			
	U_Shape	0.5573	0.5586	0.5591	0.5583			
	WWPF	0.5955	0.5969	0.5972	0.5965			
UIQM	Set	TE	VAL	TR	Average			
	None	0.8826	0.8635	0.8654	0.8705			
	DF	1.1001	1.0748	1.0752	1.0834			
	FFA	0.8826	0.8635	1.2587	1.0016			
	U_Shape	1.3459	1.3446	1.3454	1.3453			
	WWPF	1.9078	1.8887	1.8981	1.8982			

10.2 Appendix B – DockerFiles

10.2.1 CIRNet

```
FROM rocm/pytorch:rocm7.2_ubuntu22.04_py3.10_pytorch_release_2.9.1
RUN pip install --upgrade pip && \
    pip install \
        numpy \
        Pillow \
        matplotlib \
        timm \
        scipy \
        opencv-python
WORKDIR /workspace

COPY . .

RUN pip install --upgrade pip && \
    pip install \
        numpy \
        Pillow \
        matplotlib \
        timm
```

10.2.2 TC-USOD

```
FROM rocm/pytorch:rocm7.2_ubuntu22.04_py3.10_pytorch_release_2.9.1
WORKDIR /workspace

COPY . .

RUN pip install --upgrade pip && \
    pip install \
        torchvision \
        numpy \
        Pillow \
        matplotlib \
        timm
```

10.2.3 DehazeFormer

FROM rocm/pytorch:rocm7.2_ubuntu22.04_py3.10_pytorch_release_2.9.1

WORKDIR /workspace

COPY . .

```
RUN pip install --upgrade pip && \
    pip install \
        "numpy<2.0" \
        opencv-python==4.5.5.62 \
        tqdm \
        pytorch-msssim \
        timm \
        tensorboard \
        tensorboardx
```

10.2.4 U-Shape Transformer

FROM rocm/pytorch:rocm7.2_ubuntu22.04_py3.10_pytorch_release_2.9.1

WORKDIR /workspace

COPY . .

```
RUN pip install --upgrade pip && \
    pip install jupyter \
    einops \
    timm==0.4.12 \
    opencv-python \
    scikit-image \
    PyWavelets \
    tqdm \
    tensorboardX \
    munch \
    yacs \
    ml-collections
```

10.3 Appendix C – Project Costing

No external purchases were made during this project. All computation was performed on personally owned hardware. No technician contribution was required.